Surfacing boundary objects:measuring context diversity in feminist literary history

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The situatedness of knowledge (Haraway 1988) is a challenge for digital literary history: taking into account different perspectives and different modes of collecting and representing knowledge relevant to the cultural past necessarily complexifies the approach. Important computational studies have analyzed large bodies of data for evidence of change, for instance in the representation of gender, and have linked those changes to broader historical shifts (e.g. Bode 2009; Underwood 2019). Can we bring this kind of context right into our data rather than invoking it generally? Can connections that work across domains provide clues to turning points, trends, or patterns that we don't already suspect, or uncover features of the literary historical landscape that might surprise us or trouble our models and narratives of how the literary and the contextual interact? Advances in literary history, particularly recuperative or revisionary histories, often emerge from combining evidence from multiple contexts that, taken together, indicate the importance of an author, text, publisher, or other phenomenon. Within a situated model of literary history, the appearance of an entity in a variety of contexts may therefore help identify change or significance. Among the most stimulating concepts to emerge from considerations of how entities signify in multiple contexts is that of boundary objects (Star / Griesemer 1989; Star 2016). This paper reports on attempts to incorporate Susan Leigh Star's theory of boundary objects into methods of pursuing computational literary history.

Boundary objects span multiple perspectives, contexts, and communities of practice. They have strong temporal qualities, are both concrete or specific and abstracted, and have much to do with power relations and the conceptual systems that organize worlds materially and culturally. They are often heterogeneous internally, like the bodies of information and the datasets that are typical of much literary historical inquiry and indeed much inquiry in the humanities.

Our questions are: can the concept of boundary objects, grounded in an understanding of knowledge as situated, usefully inform the digital pursuit of literary history? In particular, can it advance a literary history that recognizes what Ta-Nehisi Coates calls "the political power of culture" - that is, not only formal innovation but also the shaping of lives and of historical processes? How might a computational approach incorporate an understanding of knowledge as situated and mobilize the concept of boundary objects? Approaching literary historical inquiry through boundary objects, we posit, can help to identify significant entities or trends of which we are unaware and provide perspectives on the literary past that shift received views.

This short paper reports on our efforts to locate boundary objects within a feminist literary historical dataset derived from the ongoing *Orlando: A History of Women's Writing in the British Isles from the Beginnings to the Present* (2006-) (Brown / Clements / Grundy 2024). *Orlando*'s extensive semantic dataset has been co-authored over the course of three decades by more than 100 people. It represents the lived experiences, publication histories, and literary reception of 1466 authors, along with their social, cultural, and political contexts, drawing from numerous sources and covering a wide temporal range. Although structured by a set of extensive bespoke XML markup schemas (Brown et al. 2025a; Brown et al. 2025b; Brown / Clements / Grundy 2024; 2006) and guided by editorial processes, the Orlando dataset has some of the heterogeneity and unevenness that characterize datasets drawn from disparate sources as a result of being authored by many hands

with somewhat diverse priorities, expertise, and perspectives in a blend of structured and semi-structured formats. The range of tags in the schemas—from family to politics, advertising, versification, and relationships with publishers—indicates the breadth of content.

In this analysis, we generated tabular data using Python extraction scripts that inferred connections between entities based on the nested hierarchies of the XML markup in which they occurred. The resulting network's tabular data provides an extensive overview of women's writing in Britain, with 251,834 connections between 81,639 entities (people, places, organizations, and texts). [1]

Our initial work with the *Orlando* data included reviewing the features of mainstream boundary objects and prominent examples from feminist literary history as potential means of identifying interesting entities. We found that the disparity in the amount of information available for different authors, along with the impact of previous judgements, creates a form of popularity bias that makes it difficult to identify boundary objects on the basis of interconnectedness alone.

However, Orlando's semantic queries permit both the pruning of graphs and the categorization of contexts by broad type. We investigated whether a broader range of context types would indicate a potential boundary object or entity of interest. Studies of linked information have found that diversity can be measured by how many entity types exist, the degree to which types can be differentiated, and the distribution of types (Morales et al. 2021; Stirling 2007). We devised a context diversity measure that both draws on this strategy by differentiating and analyzing the distribution of edge types to factor in the breadth of context surrounding entities, and also prioritizes diversity in the cultural sense of valuing difference. Although women writers from the British Isles are not a hugely underrepresented group relative to other women, identifying context diversity within this group has the potential to help address the popularity biases within the data and to detect writers within the cohort whose importance might have been overlooked.

Our initial experiments found that standard centrality measures such as betweenness centrality or degree centrality, both of which analyze the number of in-degree links to a node where nodes with higher counts are seen as more central (Caballera 2024), reinforce popularity bias in the data and do not assist in identifying multivalent entities. We assigned one or more contexts to each connection based on the semantic XML tags surrounding the passages in the text from which the connections between entities were derived. Leveraging the schemas, we selected 19 high-level contexts (working from tags in the dataset such as birth, health, politics, location, occupation, production, and reception) as the basis for a context diversity score which counts how many different contexts for connections are associated with each entity within the graph. The contexts were not weighted counts–each was counted only once to calculate the spread of context types associated with an entity. This first context diversity measure thus indicates the number of different edge types associated with each node in the network.

Calculating a context diversity score for each node, we compared the results with degree centrality scores. Whereas degree centrality largely reinforces received views of relative importance, reflecting the literary canon, context diversity scores revealed several boundary objects within the dataset, writers whose importance was not conventionally 'literary': Queen Victoria, feminist activist Bessie Rayner Parkes, and writers who changed the literary or social scene without necessarily distinguishing themselves for their originality as authors, such as Sylvia Beach and Emily Faithfull. Comparing a visualization based on standard measures of centrality (Fig. 1) to one foregrounds nodes based on the entity's context diversity score (Fig. 2) reveals the extent to which making entities with a greater spread of contexts more prominent provides a different perspective on the relative importance of various entities, particularly for people and texts.

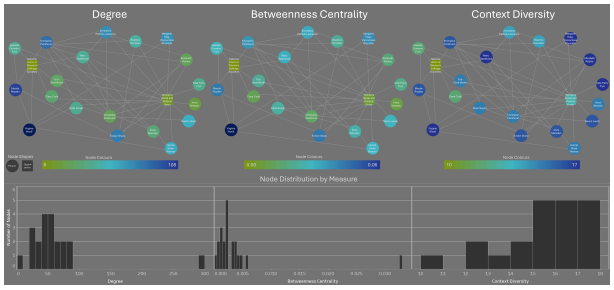

Figure 1. Graph of all entities with the 12 nodes with the highest degree centrality labelled. Node size reflects centrality. Node colour reflects betweenness.

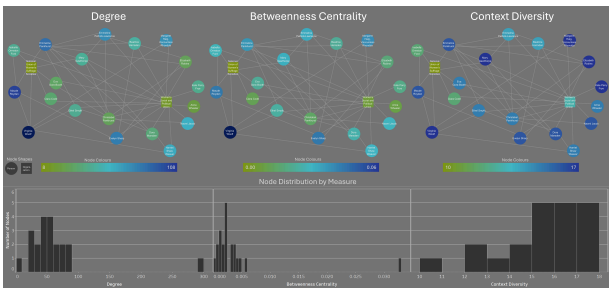

Figure 2. Graph of all entities with 12 labelled nodes from the highest 2 levels of context diversity. Node size reflects centrality. Depth of node colour reflects high context diversity scores.

These initial results support the hypothesis that seeking boundary objects by attending to context diversity within large literary historical graphs affords a different perspective on significance that complements centrality measures and can help to counter popularity bias, as shown in parallel visualizations of a small subgraph associated with the two leading UK women's suffrage organizations (Fig. 3). Our results also align with attempts to identify underrecognized authors in the *Orlando* dataset by other means (Baird 2014).

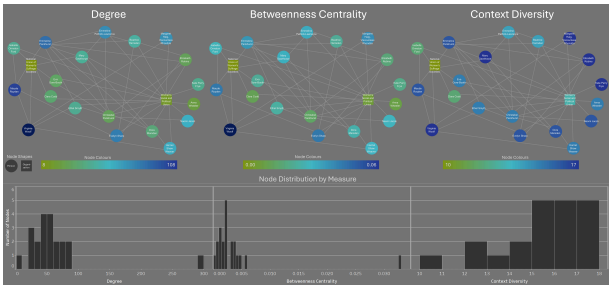

Figure 3. Subgraph of the Orlando graph shown coloured by three different network measures: degree, betweenness centrality, and context diversity. All measures are calculated from the entire Orlando graph rather than this subset showing relationships between people involved in the two most prominent women's suffrage societies in the UK. Histograms showing the distribution of measure values are shown below. Compared to the other two measures where Virginia Woolf is isolated as the largest value, context diversity indicates a wider array of nodes (Mary Gawthorpe, Margaret Haig, Elizabeth Robins, and Kate Perry Frye) that may be of interest.

These promising results indicated the desirability of refining the context diversity measure, experimenting with other means of evaluating diversity in the dataset, and of attempting to counter bias and unevenness in this kind of data. We will share the results of using revised context diversity measures that combine and differently weight context types to balance better their semantic implications, and that try to offset the extent to which context diversity partially reflects entry length, canonicity, and temporal period. We will report on experiments with algorithms designed to evaluate diversity in networks, including Kendall's Rank Correlation Coefficient and the Shannon Diversity model (Nikolov et al. 2015). Some of the bias in the data may be addressed by levaging algorithms associated with social network analysis and scale-free networks–or graphs that have a degree distribution that follows a power law and demonstrates a robustly connected network where hubs can be identified (Barabási / Albert 1999; Ciampaglia / Flammini / Menczer 2015; Salganik / Diaz / Farnadi 2024; Wood 2024).

Employing context diversity measures helps to surface boundary objects in a knowledge graph in ways that standard social network analysis measures, with their emphasis on centrality, do not. This approach could advance literary historical inquiry that combines data from a wide range of sources, although it would require adaptation, since such measures must be based on deep knowledge of both data models and content. Context diversity evaluation must emerge from an iterative hermeneutical process that will not produce "universal" measures, and will be most useful when employed with "thick" data that reflects a wide range of contexts. This kind of data is still quite rare, because meaningfully relating disparate types of connections involves significant data modeling, aligning, and linking, as well as contending with unevenness and bias. However, literary history has always spanned multiple domains. These results suggest the potential of bringing together data produced by approaches as diverse as bibliography, stylistics, and prosopography. Doing so may lead us to boundary objects that help to revise and nuance dominant historical narratives, providing fresh perspectives on the cultural past as we continue to explore ways to investigate history at scale.

Notes

1. The data analyzed here was produced from the source XML underlying the published textbase. The size, complexity of markup, and disparate data models make harmonizing and working with this data as XML challenging. Those wishing to work with the data in a graph form that documents more fully the semantic

structures can access a linked open dataset generated from the XML using the Canada Writing Research Collaboratory ontology (Brown, Cummings, et al.; Brown, Faulkner, et al.) . SPARQL queries of this dataset can extract connections between people, places, organizations, and texts (https://cwrc.ca/ https://sparql.cwrc.ca/).

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